



Module Presenter's Manual *for*

Introduction to Embedded Systems

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1. Introduction

At the end of this course, students will be able to:

- Explain what are Embedded Systems
- Explain the architecture and working of an Embedded System
- Explain the categories of Embedded Systems
- List the examples of Embedded Systems
- Explain the requirements of Embedded Systems
- Explain the future trends in Embedded Systems
- Explain the Embedded Controller architecture
- List the types of processors
- Explain the use of various peripheral devices in an Embedded Controller
- Explain the Microprocessor architecture
- Explain the Microcontroller architecture
- Explain the Digital Signal Processor architecture
- Explain the Instruction Execution cycle
- Classify the processors based on the Data Bus width
- List the difference between a Desktop Operating System and an Embedded Operating System
- Explain the Architecture of an Embedded Operating System
- List the points to be considered when selecting an operating system for Embedded Devices
- Explain the architecture of Android Operating System
- Explain the architecture of RT Linux
- List the features of VxWorks
- Explain the structure of Micro Kernel based QNX Neutrino
- Explain the architecture of Windows CE
- List the versions of Windows Embedded 7
- List the versions of Windows Embedded 8
- Explain the working of a digital circuit
- Explain the characteristics of a digital pulse
- Explain the Binary, Hexadecimal, and Decimal number system
- Perform base conversion between number systems
- Perform arithmetic operations on Binary numbers
- Explain the working of logic gates
- Perform bit masking operations on binary numbers
- Classify the microcontrollers based on their data bus width
- List the steps to select a microcontroller for embedded applications
- List the different versions of 8051 Microcontroller
- List the different vendors and their versions of the 8051 Microcontroller
- List the main features of 8051 Microcontroller
- Describe the Internal memory organization of 8051 Microcontroller
- Describe the register organization of the 8051 Microcontroller
- Explain the working of Interrupts, Timer, Input/Output ports, serial communication port in the 8051 Microcontroller
- Explain the use of power on reset
- Explain the structure of the Assembly program
- Explain the Assembler data types
- Explain the Assembler directives
- Explain the Assembly program development steps

- Create an Assembly program for an 8051 Microcontroller
- Explain how to use the Proview32 simulator
- Explain how to use the µVision simulator
- Explain the structure of the Intel Hex code
- Explain the need for data movement instructions
- List the various sources and destination locations for data Movement
- Explain the various types of instructions to move data
- Explain how to move data from external RAM and ROM
- Explain the working of a stack
- List the various arithmetic operations that can be performed in the 8051 Microcontroller
- Explain the use of Flags in arithmetic operations
- Explain how to perform addition and subtraction operations on a signed number in the 8051 Microcontroller
- List the Boolean operations supported by the 8051 Microcontroller
- Perform Boolean operations on the byte data
- List the Bit operations supported by the 8051 Microcontroller
- Perform data movement to direct addressable bits in the Internal RAM
- Perform Boolean operations on direct addressable bits
- Explain the various instructions to rotate the Accumulator bits
- Explain the steps to perform read and write operations on the I/O Ports

2. Information on Session Allocation

Module	Online Sessions (No. of Hours)
Introduction to Embedded Systems	40

Throughout this Presenter's Manual, the module **Introduction to Embedded Systems** will be referred to as **ITES**.

3. Module Deliverables

To aid the teaching process, following are the deliverables

(Faculties can access eBooks and other components in Onlinevarsity through Aptrack.)

Resources available on Onlinevarsity:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn - Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration

➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 ITES – TL1	Session 2 ITES –TL2	Session 3 ITES – TL3	Session 4 ITES – TL4
2	Session 5 ITES – TL5	Session 6 ITES – TL6	Session 7 ITES – TL7	Session 8 ITES – TL7
3	Session 9 ITES – TL7	Session 10 ITES – TL10	Session 11 ITES – TL11	Session 12 ITES – TL12
4	Session 13 ITES – TL13	Session 14 ITES – TL14	Session 15 ITES – TL15	Session 16 ITES – TL16
5	Session 17 ITES – TL17	Session 18 ITES – TL18	Session 19 ITES – TL19	Session 20 ITES – TL20

IETS: Introduction to Embedded Systems

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	IETS – TL1	All the topics as listed below from Session 1 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 1 – Introduction to Embedded Systems</u> ➤ Explain what are Embedded Systems ➤ Explain the architecture and working of an Embedded System ➤ Explain the categories of Embedded Systems ➤ List the examples of Embedded Systems ➤ Explain the requirements of Embedded Systems ➤ Explain the future trends in Embedded Systems	Introduction to Embedded Systems SG – Session 1 XP – Session 1 TG – Session 1
2	IETS – TL2	All the topics as listed below from Session 2 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 2 – Embedded Controllers</u> ➤ Explain the Embedded Controller architecture ➤ List the types of processors ➤ Explain the use of various peripheral devices in an Embedded Controller ➤ Explain the Microprocessor architecture ➤ Explain the Microcontroller architecture ➤ Explain the Digital Signal Processor architecture ➤ Explain the Instruction Execution cycle ➤ Classify the processors based on the Data Bus width	Introduction to Embedded Systems SG – Session 2 XP – Session 2 TG – Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
3	IETS – TL3	All the topics as listed below from Session 3 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 3 – Embedded Operating System</u> ➤ List the difference between a Desktop Operating System and an Embedded Operating System ➤ Explain the Architecture of an Embedded Operating System ➤ List the points to be considered when selecting an operating system for Embedded Devices ➤ Explain the architecture of Android Operating System ➤ Explain the architecture of RT Linux ➤ List the features of VxWorks ➤ Explain the structure of Micro Kernel based QNX Neutrino ➤ Explain the architecture of Windows CE ➤ List the versions of Windows Embedded 7	Introduction to Embedded Systems SG – Session 3 XP – Session 3 TG – Session 3
4	IETS – TL4	All the topics as listed below from Session 4 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 4 – Number System and Data Operations</u> ➤ Explain the working of a digital circuit ➤ Explain the characteristics of a digital pulse ➤ Explain the Binary, Hexadecimal, and Decimal number system ➤ Perform base conversion between number systems ➤ Perform arithmetic operations on Binary numbers ➤ Explain the working of logic gates ➤ Perform bit masking	Introduction to Embedded Systems SG – Session 4 XP – Session 4 TG – Session 4

Session No.	Session Title	Session Details	Deliverables' Mapping
		operations on binary numbers	
5	IETS – TL5	All the topics as listed below from Session 5 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 5 – 8051 Microcontroller Architecture</u> ➤ Classify the microcontrollers based on their data bus width ➤ List the steps to select a microcontroller for embedded applications ➤ List the different versions of 8051 Microcontroller ➤ List the different vendors and their versions of the 8051 Microcontroller ➤ List the main features of 8051 Microcontroller ➤ Describe the Internal memory organization of 8051 Microcontroller ➤ Describe the register organization of the 8051 Microcontroller ➤ Explain the working of Interrupts, Timer, Input/Output ports, and serial communication port in the 8051 Microcontroller ➤ Explain the use of power on reset in the 8051 Microcontroller	Introduction to Embedded Systems SG – Session 5 XP – Session 5 TG – Session 5
6	IETS – TL6	All the topics as listed below from Session 6 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 6 – 8051 Assembly Programming</u> ➤ Explain the structure of the Assembly program ➤ Explain the Assembler data types ➤ Explain the Assembler	Introduction to Embedded Systems SG – Session 6 XP – Session 6 TG – Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		directives ➤ Explain the Assembly program development steps ➤ Create an Assembly program for an 8051 Microcontroller ➤ Explain how to use the Proview32 simulator ➤ Explain how to use the µVision simulator ➤ Explain the structure of the Intel Hex code	
7	IETS – TL7	All the topics as listed below from Session 7 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 7 – Data Movement and Arithmetic Operations</u> ➤ Explain the need for data movement instructions ➤ List the various source and destination locations for data movement ➤ Explain the various types of instructions to move data ➤ Explain how to move data from external RAM and ROM ➤ Explain the working of a stack	Introduction to Embedded Systems Introduction to Embedded Systems SG – Session 7 XP – Session 7 TG – Session 7
8	IETS – TL8	All the topics as listed below from Session 7 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 7 – Data Movement and Arithmetic Operations</u> ➤ List the various arithmetic operations that can be performed in the 8051 Microcontroller ➤ Explain the use of Flags in arithmetic operations ➤ Explain how to perform addition and subtraction operations on a signed number	Introduction to Embedded Systems SG – Session 7 XP – Session 7 TG – Session 7
9	IETS – TL9	All the topics as listed below from Session 8 of <i>Introduction to Embedded Systems</i> book should	Introduction to Embedded Systems

Session No.	Session Title	Session Details	Deliverables' Mapping
		be covered in this session. <u>Session 8 – Bitwise and Boolean Operations</u> ➤ List the Boolean operations supported by the 8051 Microcontroller ➤ Perform Boolean operations on the byte data ➤ List the Bit operations supported by the 8051 Microcontroller	SG – Session 8 XP – Session 8 TG – Session 8
10	IETS – TL10	All the topics as listed below from Session 8 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 8 – Bitwise and Boolean Operations</u> ➤ Perform data movement to direct addressable bits in the Internal RAM ➤ Perform Boolean operations on direct addressable bits ➤ Explain the various instructions to rotate the Accumulator bits ➤ Explain the steps to perform read and write operations on the I/O Ports	Introduction to Embedded Systems SG – Session 8 XP – Session 8 TG – Session 8
11	IETS – TL11	All the topics as listed below from Session 9 of <i>Introduction to Embedded Systems</i> book should be covered in this session. <u>Session 9 – Branching Operations</u> ➤ Explain Conditional and Unconditional transfer of control ➤ Explain the address range in the branching instructions ➤ Explain the working of bit conditional jumps ➤ Explain the working of byte conditional jumps ➤ Explain the working of short,	Introduction to Embedded Systems Introduction to Embedded Systems SG – Session 9 XP – Session 9 TG – Session 9

Session No.	Session Title	Session Details	Deliverables' Mapping
		absolute, and long jumps ➤ Explain the working of delay loops	
12	IETS – TL12	All the topics as listed below from Session 10 of Introduction to Embedded Systems book should be covered in this session <u>Session 10 – Subroutines and Interrupts</u> ➤ Explain the importance of subroutines in Embedded Controllers ➤ Explain the working of the subroutines in the 8051 Microcontroller ➤ Explain how to create and call the subroutines in the assembly code	Introduction to Embedded Systems Session 10 Session 10 Session 10
13	IETS – TL13	All the topics as listed below from Session 10 of Introduction to Embedded Systems book should be covered in this session <u>Session 10 – Subroutines and Interrupts</u> ➤ Explain the working of interrupts in the 8051 Microcontroller ➤ Explain how to use the timers in the 8051 Microcontroller ➤ Explain how to use the timers to generate delays for fixed time period ➤ Explain how to use the timers as counters in the 8051 Microcontroller	Introduction to Embedded Systems Session 10 Session 10 Session 10
14	IETS – TL14	All the topics as listed below from Session 11 of Introduction to Embedded Systems book should be covered in this session <u>Session 11 – Developing Efficient Assembly Programs</u> ➤ Explain the use of Pseudocodes to develop program logic	Introduction to Embedded Systems Session 11 Session 11 Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain the Pseudocode building blocks ➤ Explain various Addressing Modes and their usage ➤ Explain the steps to optimize the Assembly Program	
15	IETS – TL15	All the topics as listed below from Lab Session 1 of Introduction to Embedded Systems book should be covered in this session <u>Session 1 – 8051 Assembly Programming</u> ➤ Use µVision Simulator to create, assemble, and execute an 8051 Assembly program	Introduction to Embedded Systems (LAB) Session 1
16	IETS – TL16	All the topics as listed below from Lab Session 2 of Introduction to Embedded Systems book should be covered in this session <u>Session 2 – Data Movement and Airthmetic Operations</u> ➤ Create a program to move data to register banks 1, 2, and 3 ➤ Create a program to move data to and from Internal RAM ➤ Create a program to add two 16-bit numbers ➤ Create a program to subtract two numbers ➤ Create a program to multiply two numbers ➤ Create a program to divide two numbers	Introduction to Embedded Systems (LAB) Session 2
17	IETS – TL17	All the topics as listed below from Lab Session 3 of Introduction to Embedded Systems book should be covered in this session <u>Session 3 – Bitwise and Boolean Operations</u> ➤ Create a program to read data	Introduction to Embedded Systems (LAB) Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		from a port pin and output its complement on another pin ➤ Create a program to read a port bit and store it in the bit addressable RAM location ➤ Create a program to clear a port bit using the AND operation ➤ Create a program to move the MSB of the Accumulator to a port pin	
18	IETS – TL18	All the topics as listed below from Lab Session 4 of Introduction to Embedded Systems book should be covered in this session <u>Session 4 – Branching Operations</u> ➤ Create a program to find the largest of 10 numbers stored in the RAM at location 30H ➤ Create a program to generate a delay of 1 second	Introduction to Embedded Systems (LAB) Session 4
19	IETS – TL19	All the topics as listed below from Lab Session 5 of Introduction to Embedded Systems book should be covered in this session <u>Session 5 – Subroutine and Interrupts</u> ➤ Create a program using subroutine to calculate factorial of a number ➤ Create a program to use the 8051 timer to create a delay of 5 milliseconds ➤ Create a program to program the timer ISR to generate an alert	Introduction to Embedded Systems (LAB) Session 5
20	IETS – TL20	All the topics as listed below from Lab Session 6 of Introduction to Embedded Systems book should be covered in this session <u>Session 6 – Developing Efficient Assembly Program</u>	Introduction to Embedded Systems (LAB) Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Create a program to move data from code memory to External Data memory ➤ Create a program to move data from code memory to Internal RAM ➤ Create a program to move data from External Data memory to Internal RAM ➤ Create a program to move data from Internal RAM to External Data memory	

6. Library References

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| ➤ Introduction to Embedded Systems: A Cyber-Physical Systems Approach
Second Edition — MIT Press — 2017 by Lee and Seshia |
| ➤ The Art of Designing Embedded Systems - 2nd Edition by Jack Ganssle |

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